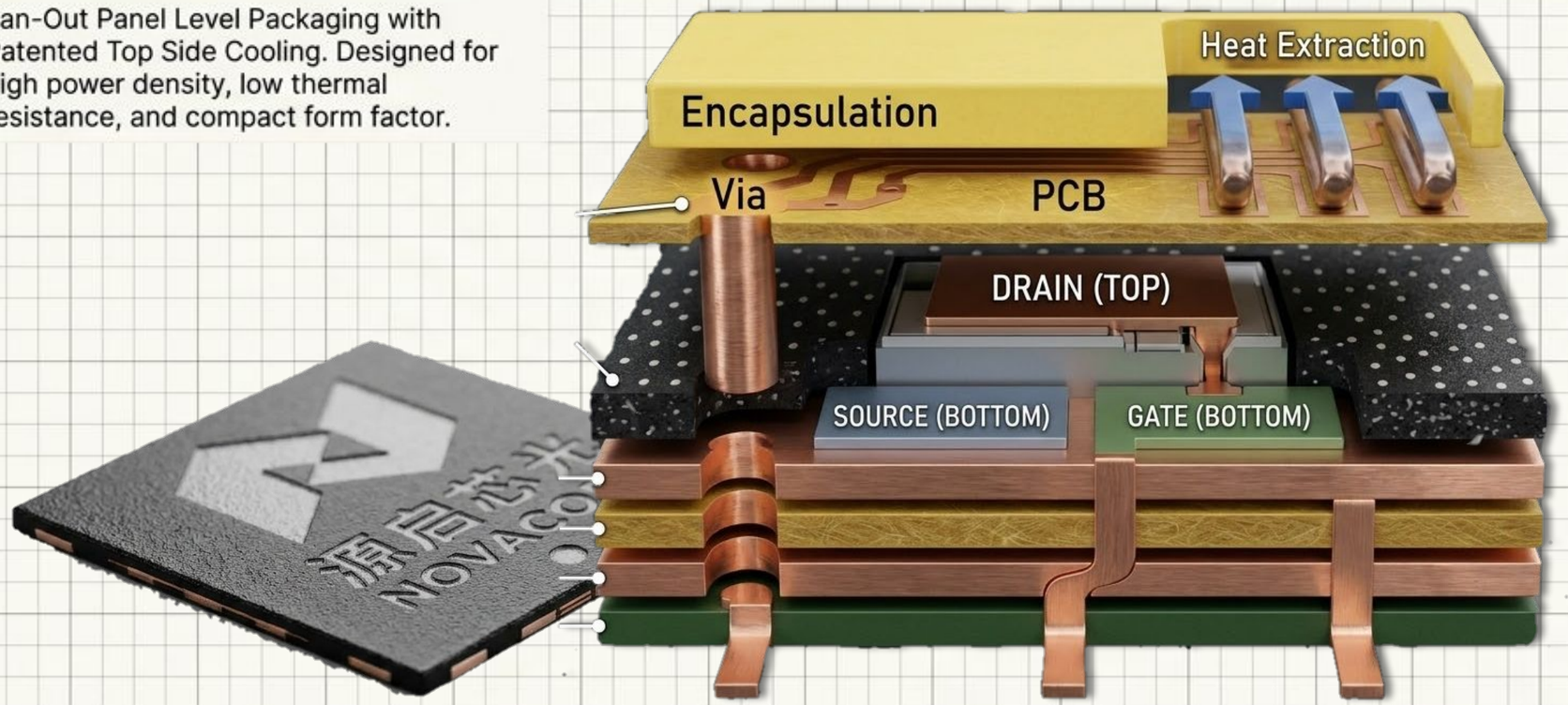
NOVACORA Single N-MOSFET FOPLP+TSC Specification

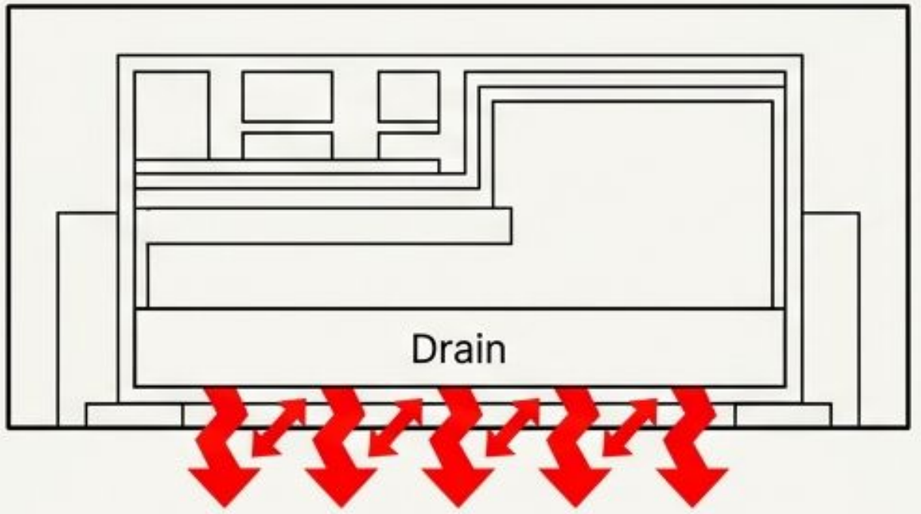
Fan-Out Panel Level Packaging with Patented Top Side Cooling. Designed for high power density, low thermal resistance, and compact form factor.



DOCUMENT: Internal Technical Reference
REVISION: October 2026
STATUS: Mass Production Architecture

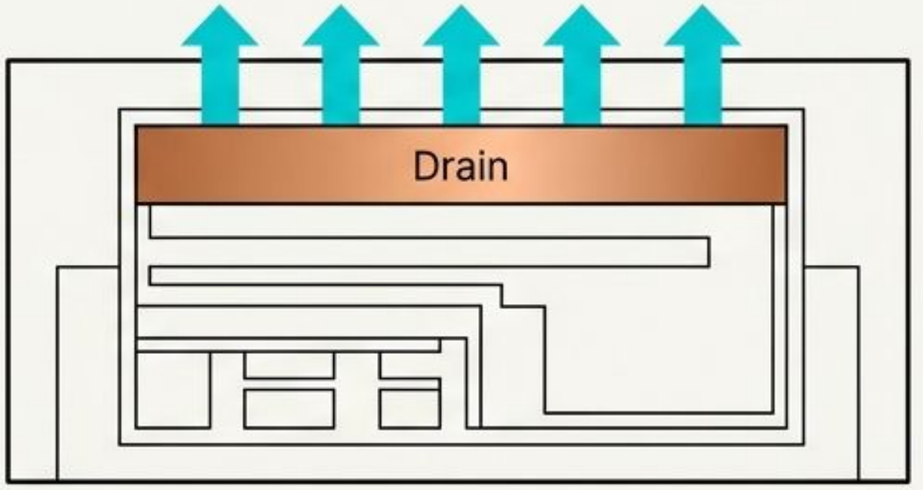
The paradigm shift of the flipped die orientation

Conventional Architecture



Standard die orientation traps heat. Drain faces the PCB, creating thermal bottlenecks in high-density applications.

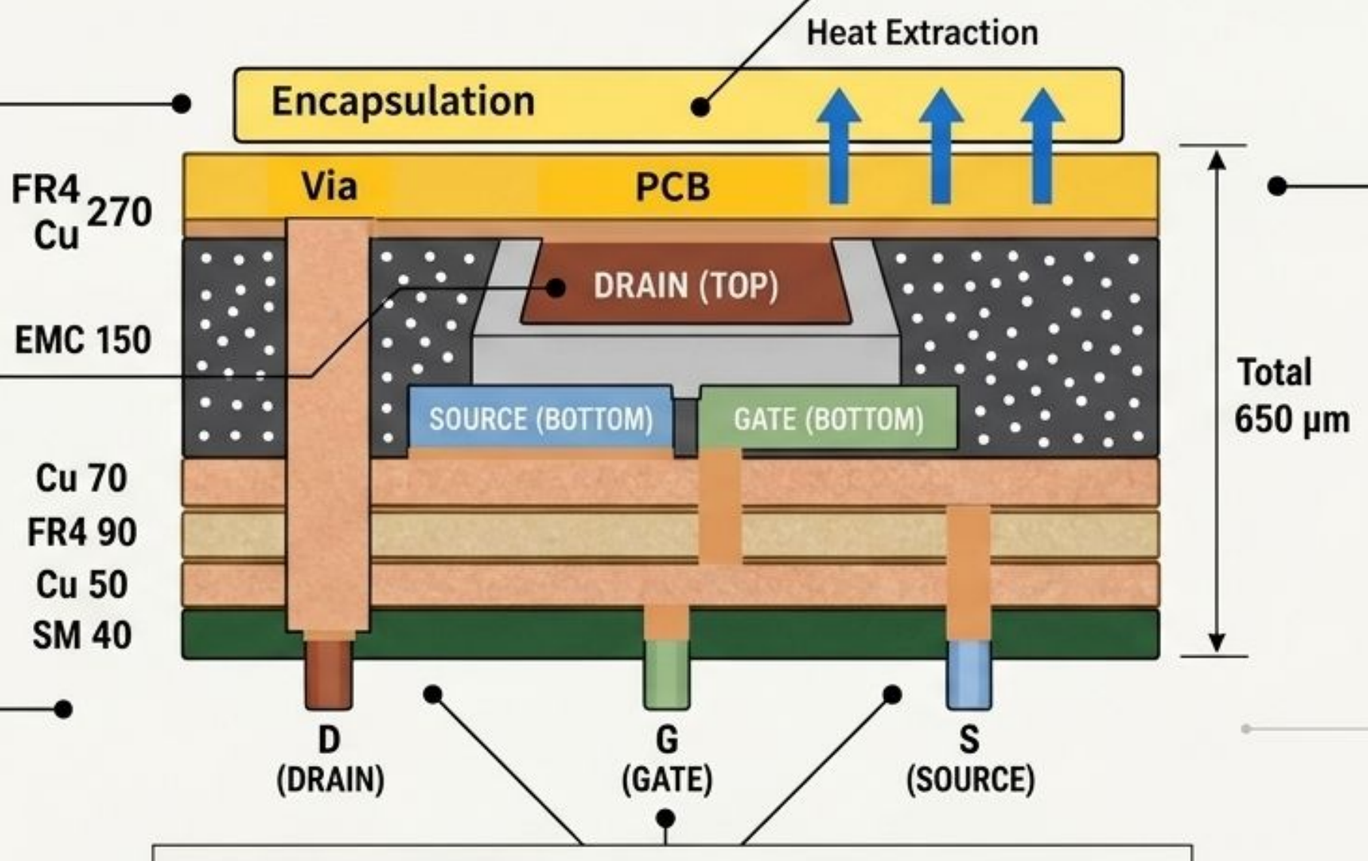
NOVACORA Paradigm



- **The Core Innovation:** The MOSFET die is embedded upside-down within the epoxy molding compound.
- **The Result:** The large-area backside metal (Drain) now faces the top of the package.
- **The Advantage:** Creates an **ultra-short, direct thermal path** for top-side heat extraction while preserving bottom SMT capability.

Unpacking the 650µm micro-architecture

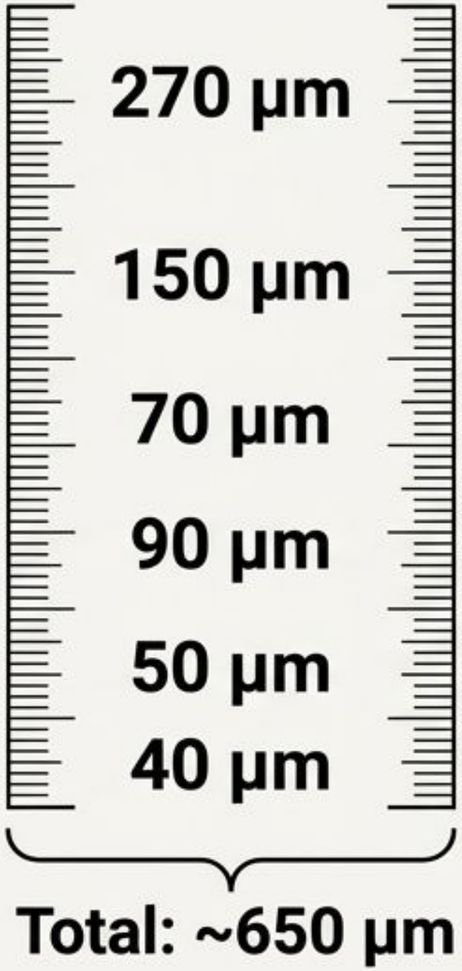
Top Side Cooling (TSC) Path
Top FR4 PCB and Encapsulation layer dedicated entirely to direct upward heat extraction.



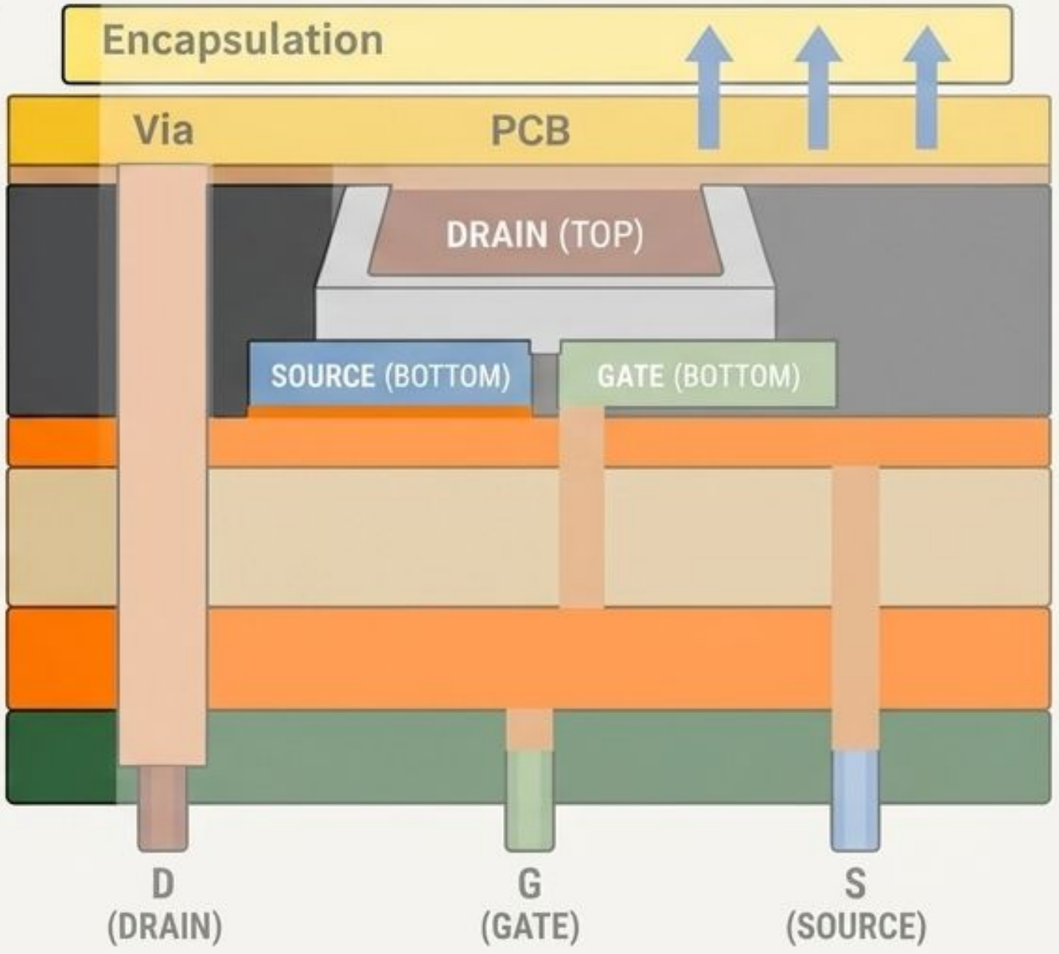
Encapsulated Flipped Die
The N-MOSFET embedded in 150µm EMC, with the large-area DRAIN specifically oriented to the top.

Bottom-Side SMT Pins
Gate (G), Source (S), and Drain (D) pins remain accessible on the bottom for standard surface-mount assembly.

Precise material distribution across the 650µm stack



PCB (FR4 + Cu)	Single copper layer for heat extraction, primary thermal spread, and top drain connection.
EMC	Epoxy Molding Compound for die encapsulation and mechanical support.
RDL (Cu)	Upper copper redistribution layer for internal routing.
FR4	Intermediate dielectric substrate.
RDL (Cu)	Bottom copper layer forming the SMT pins.
SM	Sheet Molding acting as the final solder mask and protective layer.



Upward thermal extraction through the drain layer

01 | **Generation**

Conduction loss heat originates in the channel and drift region of the die.

02 | **Primary Path**

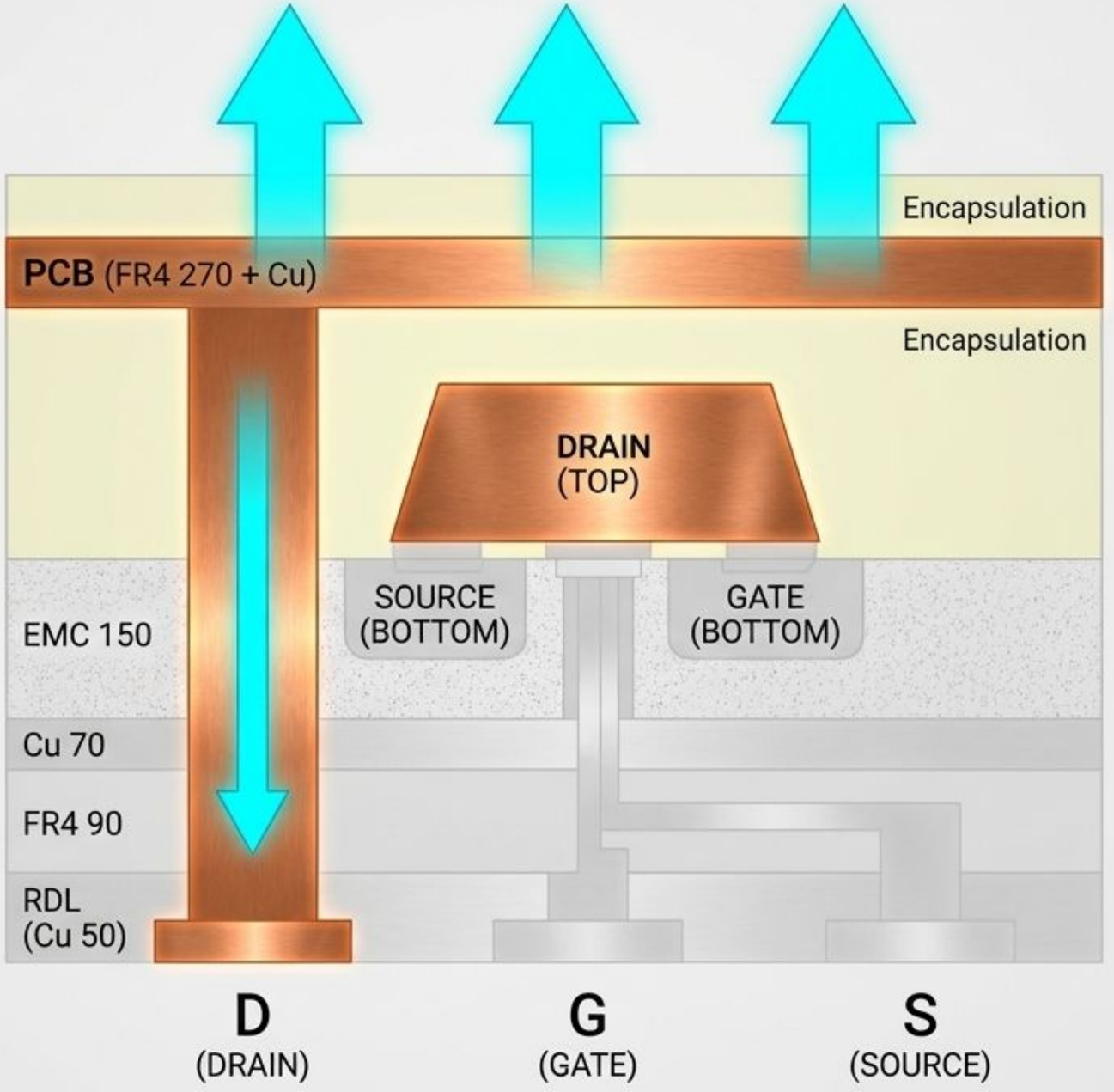
Driven upward through the large-area drain metal directly into the 270μm top FR4 RDL, exiting into the external heat sink.

03 | **Secondary Path**

Parallel heat conduction downward through the solid copper-filled vertical vias.

04 | **Result**

Drastically lower junction-to-case thermal resistance ($R_{th(j-c)}$) compared to standard packages.



Downward electrical routing via solid copper vias

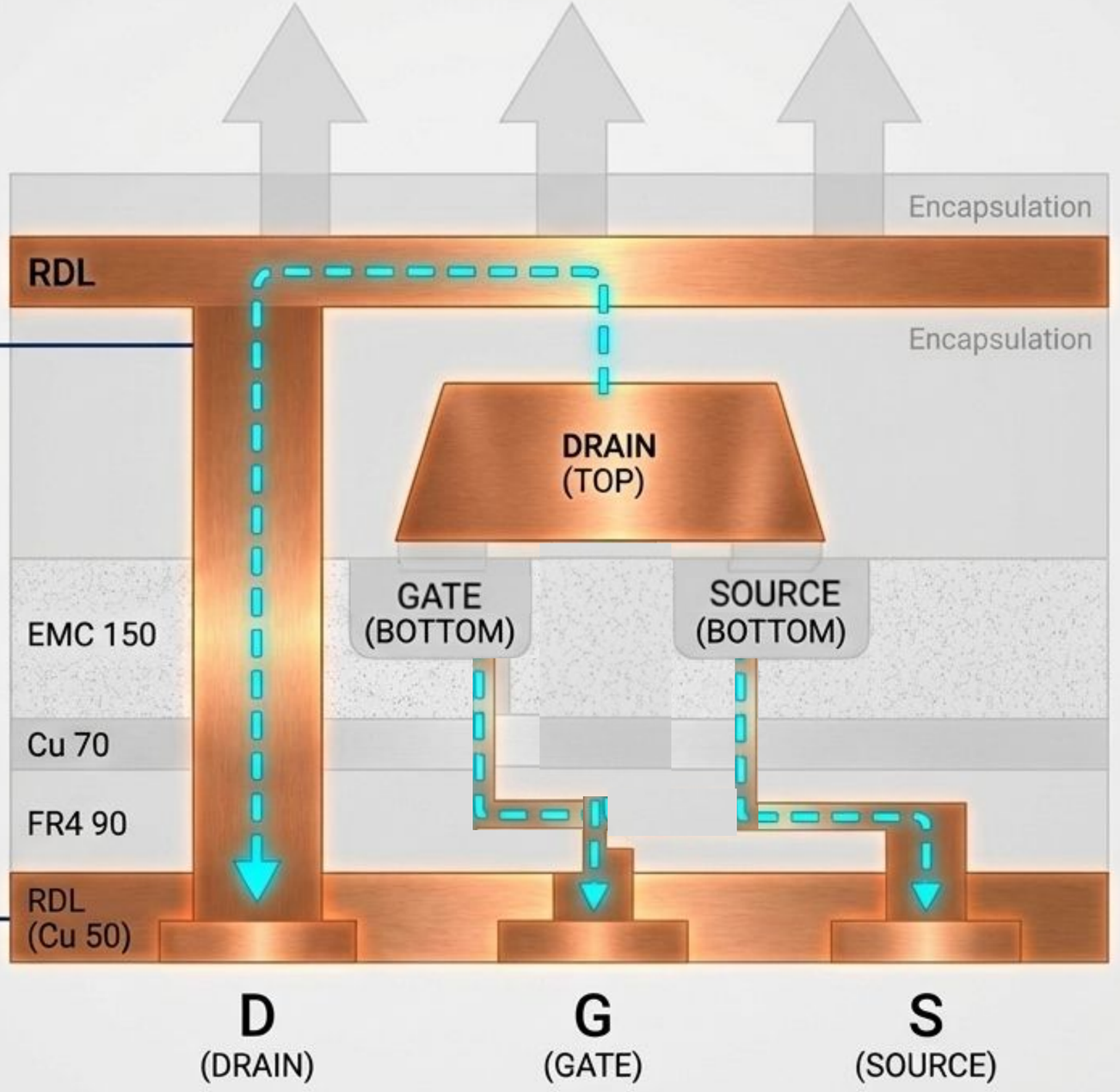
Drain (D) Routing

Top-side drain metal connects to upper Cu RDL, routing vertically downward through completely solid, void-free copper-filled vias to the bottom pin.

Gate (G) & Source (S) Routing

Located on the bottom of the flipped die, connecting directly through bottom Cu RDL layers to their respective pins.

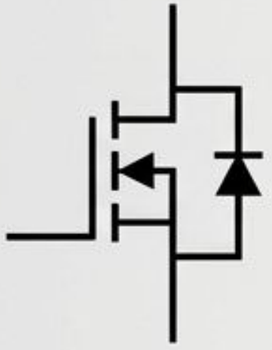
Takeaway: 100% surface-mount compatible on the bottom layer without sacrificing top-side thermal performance.



Gemini Notebook

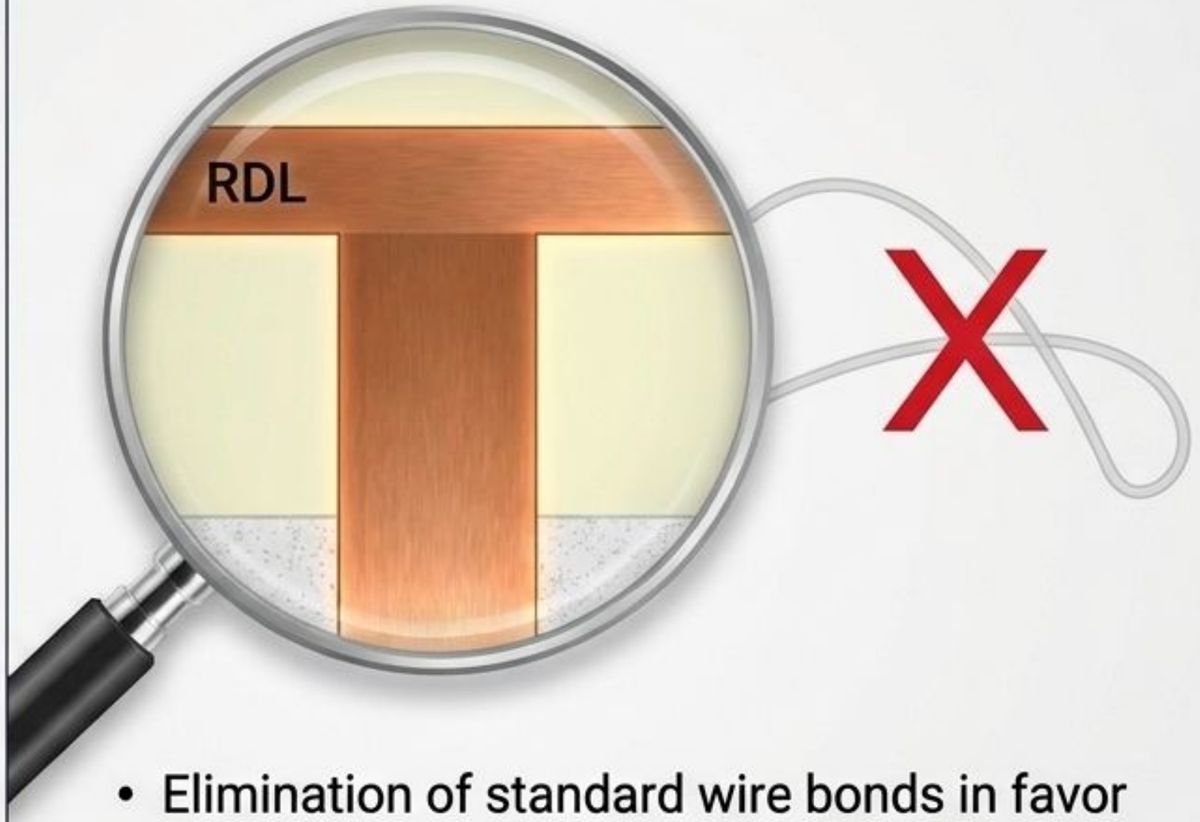
Physical package impact on operational parasitics

Standard Electrical Behavior



- **Off-state (V_{GS}=0):** Channel pinched off; leakage unchanged by package.
- **On-state (V_{GS}>V_{th}):** Conductive inversion channel forms; conduction loss dissipates via, channel heat forms; conduction loss dissipates via the Top Side Cooling path.
- The logic remains completely standard.

The Packaging Advantage

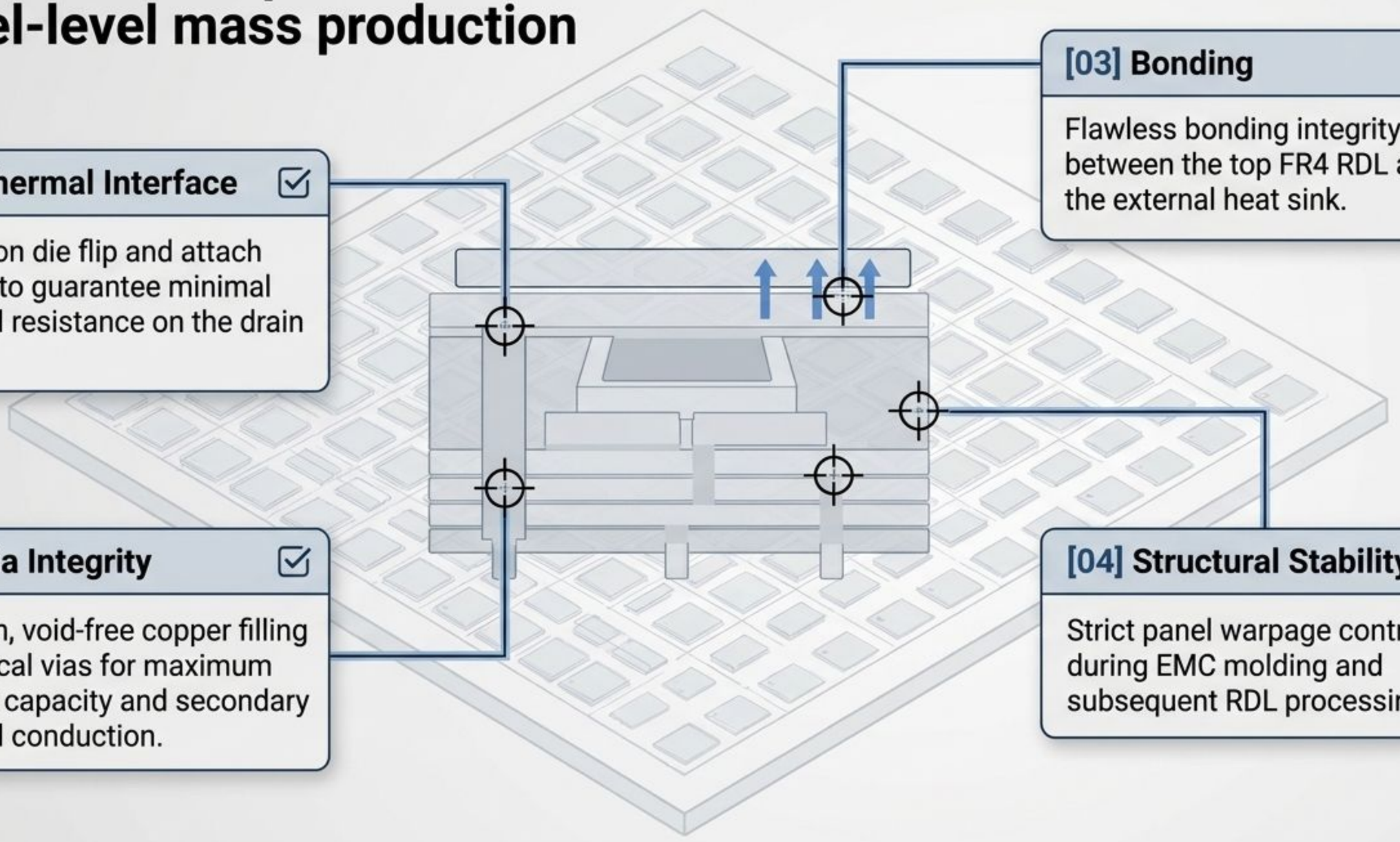


- Elimination of standard wire bonds in favor of low-inductance copper RDL and solid filled-via interconnects.
- Minimizes parasitic inductance and resistance, directly enabling higher switching frequencies and lower overall losses.

Comparing FOPLP+TSC against conventional architectures

Architectural Dimension	Conventional Package	NOVACORA FOPLP+TSC
Die Orientation	Drain Down (Standard)	**Drain Up** (Flipped)
Cooling Path	Bottom-cooled (Trapped heat)	**Direct Top Side Cooling** (TSC)
Internal Interconnects	Wire-bonded G/S	**Copper** RDL & Solid Cu Vias
Performance Impact	Higher $R_{th(j-c)}$, Higher parasitics	Minimized $R_{th(j-c)}$, Low parasitics, high frequency support

Critical control points for panel-level mass production



[01] Thermal Interface

Precision die flip and attach quality to guarantee minimal thermal resistance on the drain side.

[02] Via Integrity

Uniform, void-free copper filling of vertical vias for maximum current capacity and secondary thermal conduction.

[03] Bonding

Flawless bonding integrity between the top FR4 RDL and the external heat sink.

[04] Structural Stability

Strict panel warpage control during EMC molding and subsequent RDL processing.

Synthesizing structural choices into operational advantages



Thermal Supremacy

Feature: Large-area drain-to-heat-sink connection

The absolute lowest possible $R_{th(j-c)}$ for high-power-density applications, efficiently routing heat away from the die.



Extreme Density

Feature: ~650µm total package thickness

An ultra-thin, highly compact form factor ideal for integration into next-generation, space-constrained power modules.



Seamless Integration

Feature: Panel-level manufacturing + bottom SMT pins

Achieves high manufacturing throughput without forcing changes to conventional surface-mount assembly processes.